MASTER'S DEGREE IN PHARMACEUTICAL BIOTECHNOLOGY

From Genome to Drug – Exploring Bioinformatics & Omics Sciences

Lesson IV - Transcriptomics Analysis

Transcriptomics – Functional Enrichment

Estevão Barcelos, PostDoc UNIPG

estevaocarlos.silvabarcelos@collaboratori.unipg.it

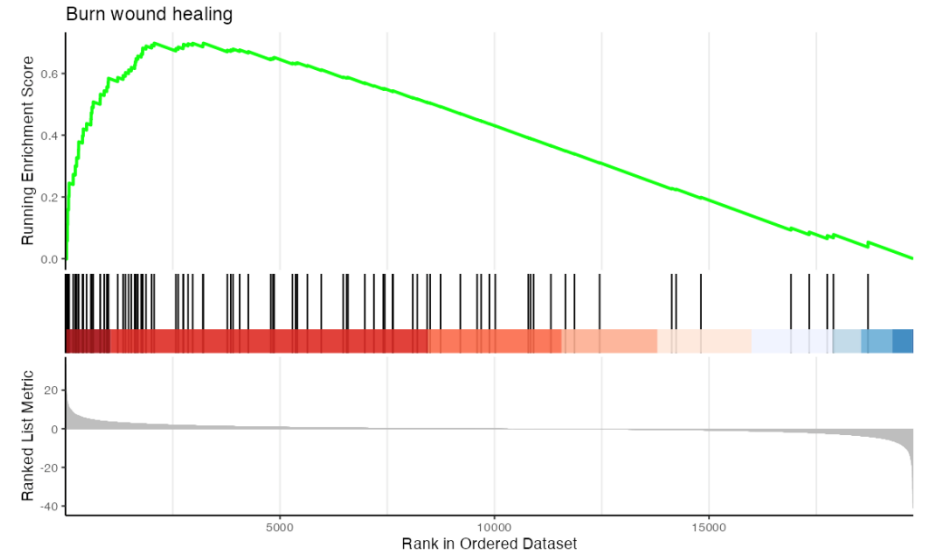
Learning Objectives — Lesson IV

What do the p-value, log2 fold change (log2FC), and adjusted p-value mean in RNA-seq?

Principles and application of Gene Set Enrichment Analysis (GSEA)

Basic principles of DESeq2

Principles and application of Gene Ontology (GO) analysis

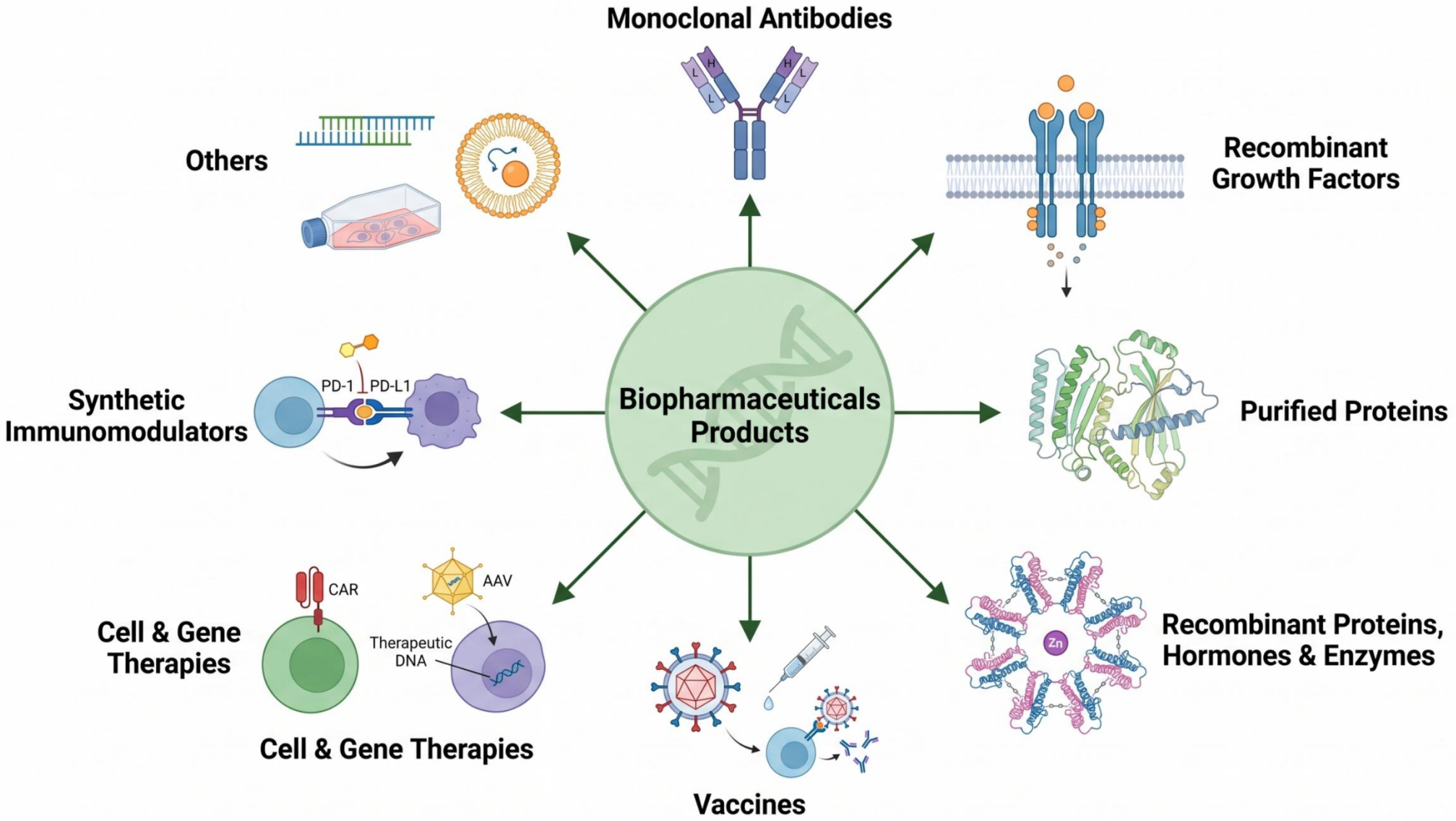




“How does a biopharmaceutical reshape gene expression?”

“Transcriptomics gives us a genome-wide snapshot of which genes are active, dysregulated, or silenced in disease”

Biopharmaceuticals Products



Main applications of Transcriptomics in Biopharmaceuticals

1

Cell Line Optimisation

Identify genes linked to productivity, growth and secretion

CHO cells: high vs low mAb productivity

2

Production Conditions

How temperature, pH and nutrients alter gene expression

Transcriptional response to temperature shift

3

Protein Quality & Processing

Protein folding, ER stress and secretion pathways

ER stress linked to reduced recombinant protein secretion

4

Post-translational Modifications

Glycosylation pathways and sugar nucleotide metabolism

Glycan composition affects antibody function

5

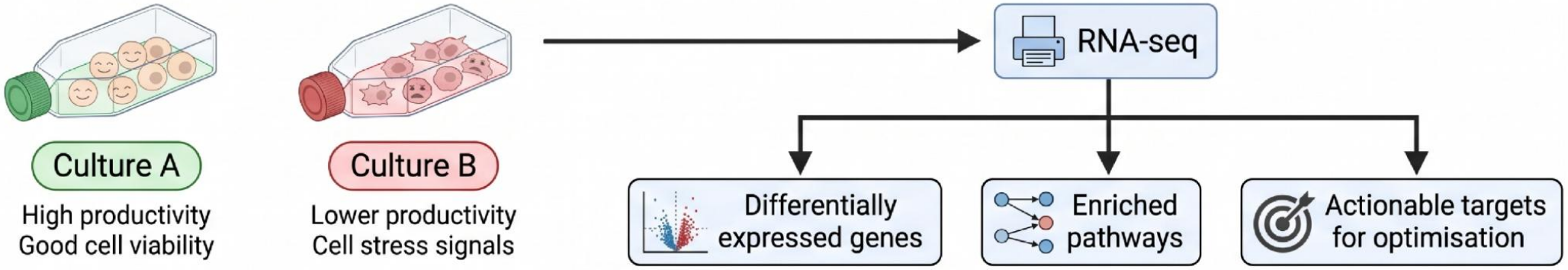
Process Consistency

Compare expression across batches, passages and conditions

Detect transcriptional signature of productivity loss

Practical Example: CHO Cell RNA-seq

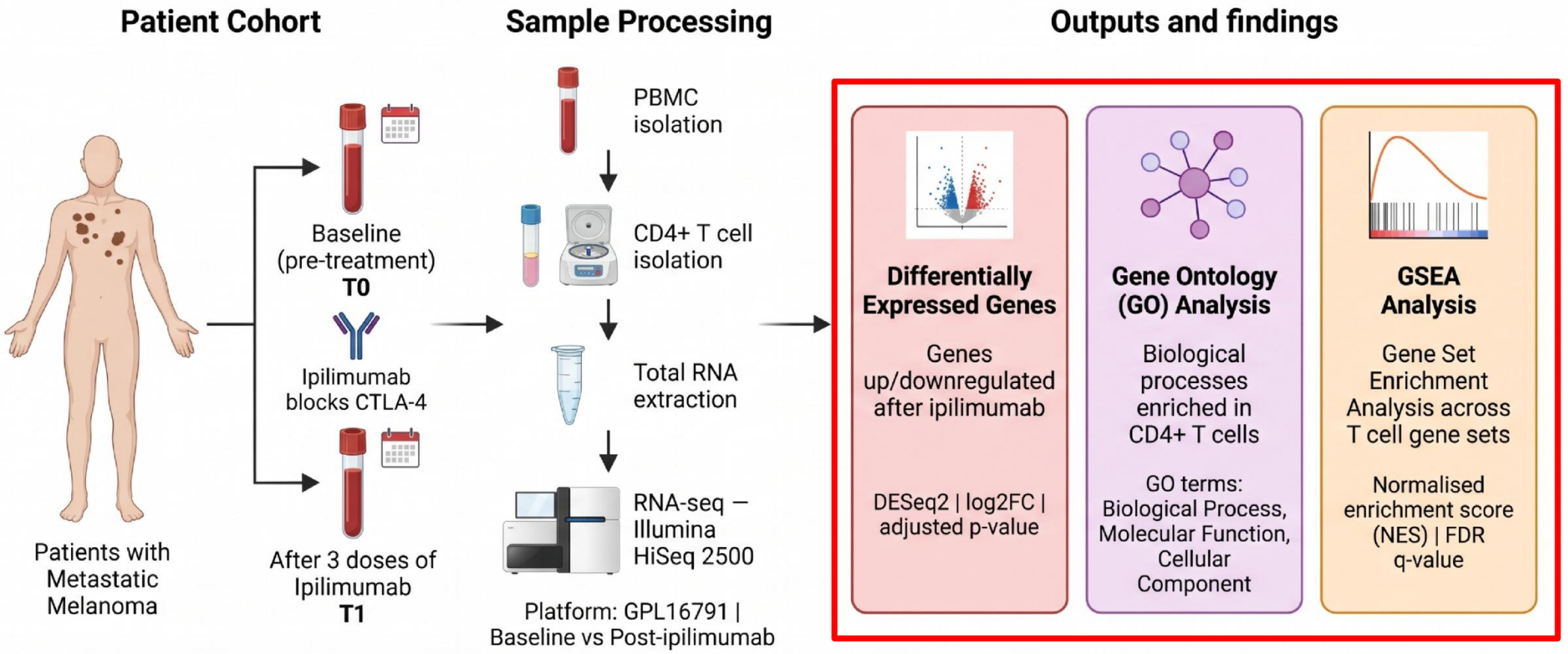
A company produces a monoclonal antibody in CHO cells and observes two phenomena



Transcriptomics in Biopharmaceutical Production

RNA-seq Analysis of CD4+ T cells following Ipilimumab Therapy

Goswami et al., 2018 — Journal of Clinical Investigation



Goswami et al. (2018) JCI | GSE114716 | Illumina HiSeq 2500 | Homo sapiens — CD4+ T cells from PBMCs



From Raw Counts to Meaningful Data

Theory behind DESeq2

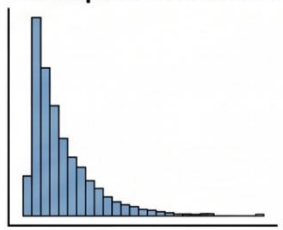
DESeq2 Workflow: Negative Binomial GLM for RNA-seq Differential Expression Analysis

The Problem: Count Data from RNA-seq

Raw Count Matrix

	Ctrl1	Ctrl2	Ctrl3	Trt1	Trt2	Trt3
Gene A	0	5	234	1500	87	92
Gene B	0	5	234	500	87	92
Gene C	0	5	198	290	67	48
...	...	...	...	...	...	...
Gene N	0	1	46	33	10	14

Overdispersed Distribution

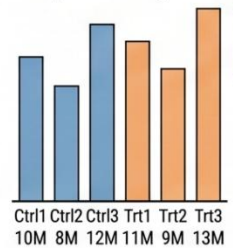


Example counts: 0, 5, 234, 1500

Raw counts are overdispersed integers — cannot use t-test or ANOVA

Step 1: Normalization by Size Factors

Unequal Library Sizes

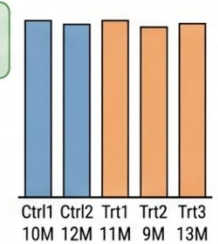


Total Reads

$$s_j = \frac{\text{median}(K_{ij})}{\text{geometric_mean}(K_i)}$$

Corrects for technical variation in sequencing depth and composition

Normalized Counts

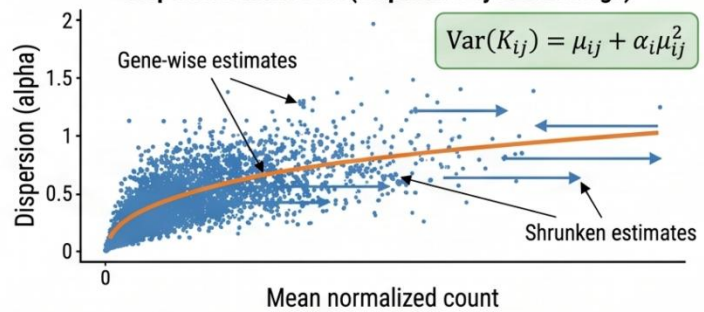


Adjusted Total Reads

Median of ratios normalization — accounts for library size AND composition bias

Step 2: Dispersion Estimation

Dispersion-Mean Plot (Empirical Bayes Shrinkage)



Negative Binomial model — captures overdispersion in count data. Empirical Bayes shrinkage pools information across genes

Step 3: Negative Binomial GLM Fitting

$$K_{ij} \sim \text{NB}(\mu_{ij}, \alpha_i)$$
$$\log(\mu_{ij}) = \sum x_{jr}\beta_{ir}$$

Design Matrix (X)

	Intercept	Condition
Ctrl1	1	0
Ctrl2	1	0
Ctrl3	1	0
Trt1	1	1
Trt2	1	1
Trt3	1	1

Coefficient Estimates

$$\begin{bmatrix} \text{Intercept } \beta_0 \\ \text{Condition } \beta_1 \end{bmatrix} \rightarrow \text{log2 fold change}$$

Generalized Linear Model with log link function — estimates log2 fold change for each gene

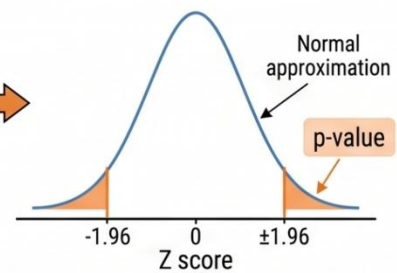
Step 4: Wald Test for Significance

Wald Test Hypotheses

H0: β = 0
H1: β ≠ 0

$$W = \frac{\hat{\beta}}{\text{SE}(\hat{\beta})}$$

Wald Statistic Distribution Under H0



Wald test: tests if log2FC is significantly different from zero. Alternative: LRT (Likelihood Ratio Test) for complex designs

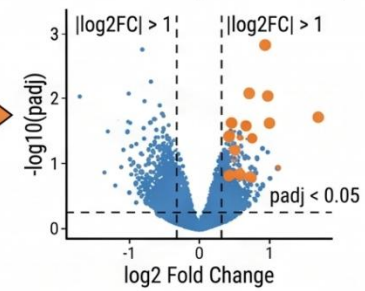
Step 5: Multiple Testing Correction

20,000 genes tested → ~1,000 expected false positives at p<0.05 by chance

p-values
Rank p-values
Compute Adjusted p-values
padj

$$padj = p \times \frac{m}{rank}$$

Volcano Plot (BH Correction)



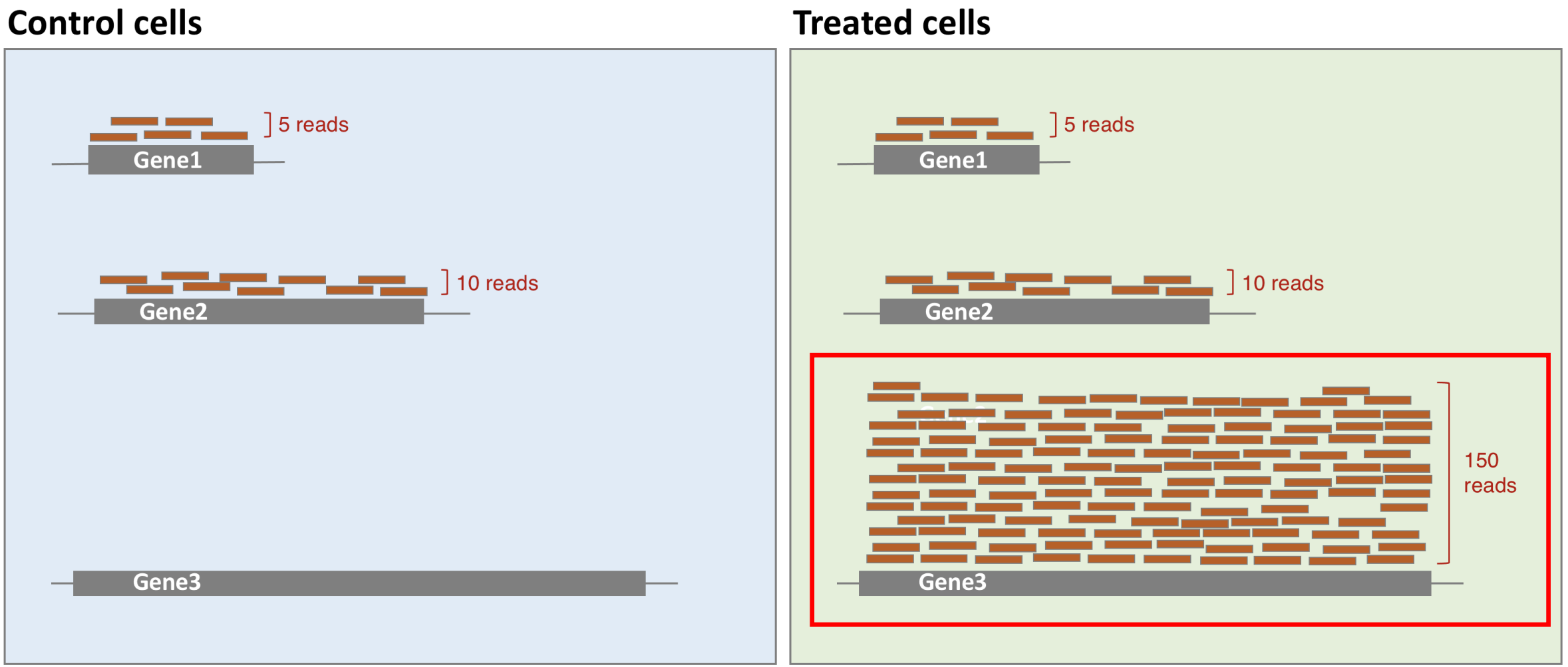
BH correction controls False Discovery Rate (FDR). Adjusted p-value (padj) is the key output

Key assumptions

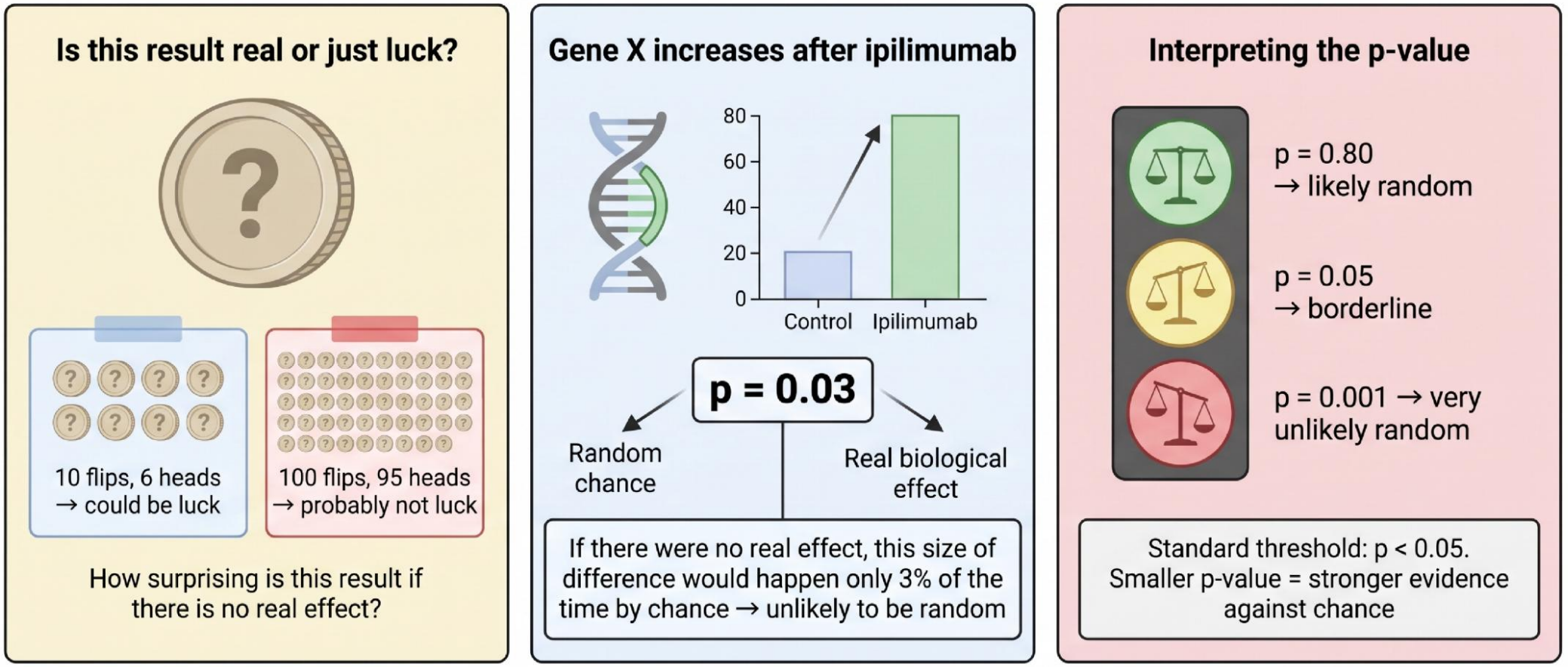
1. Counts follow Negative Binomial distribution.
2. Size factors capture systematic technical variation.
3. Dispersion is shared across similar genes.
4. Sufficient replicates per condition (minimum n=3).

In short, what do we want to see?

Let's assume that we have already normalized samples.

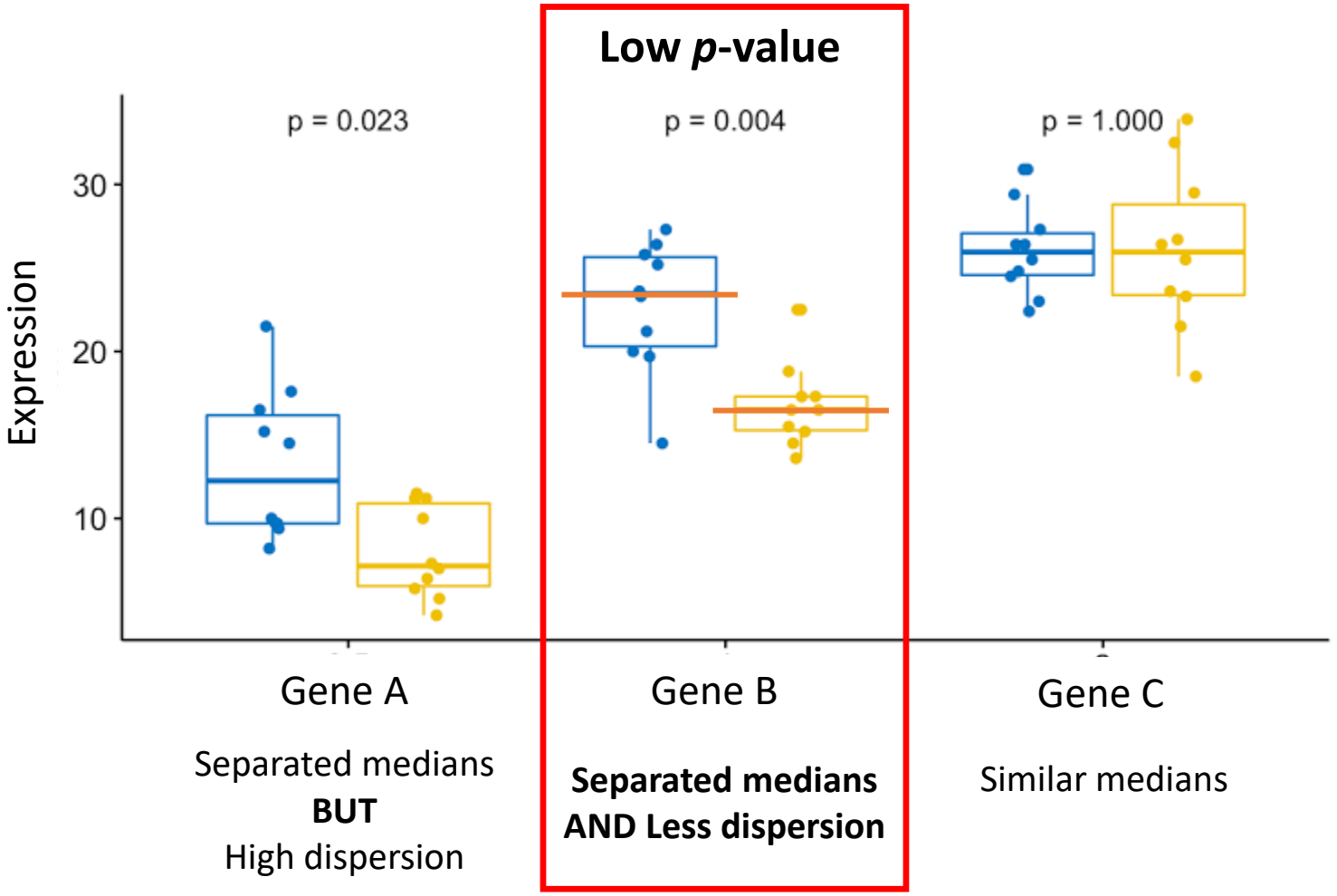


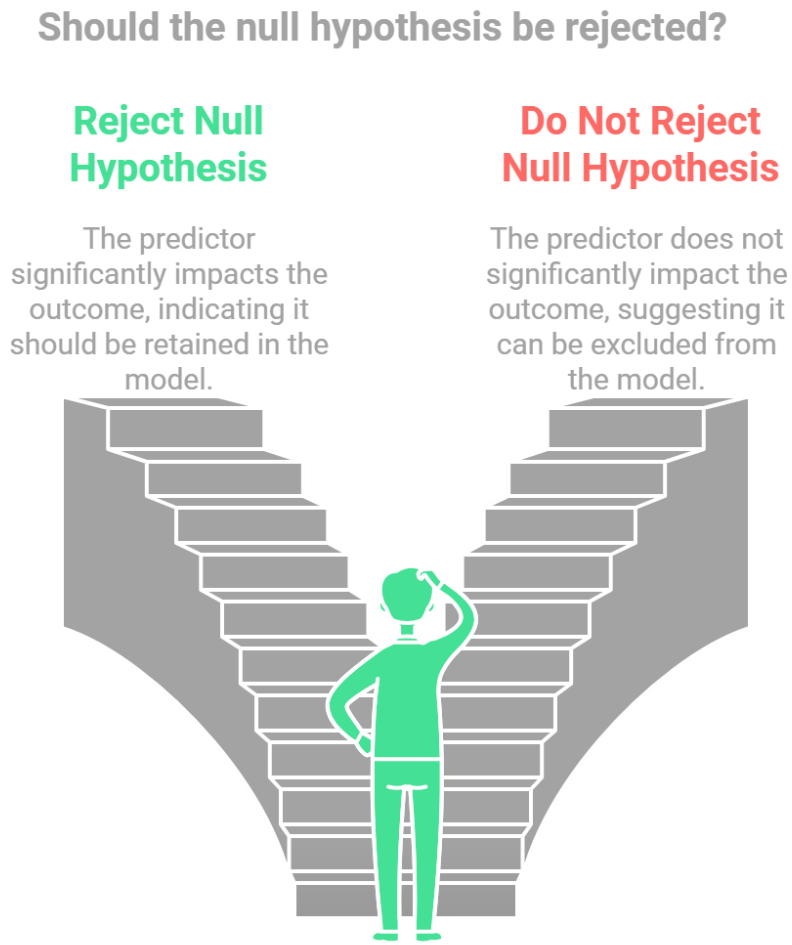
Is this result real or just luck?



The p-value does NOT measure effect size — only how likely the result is to be chance

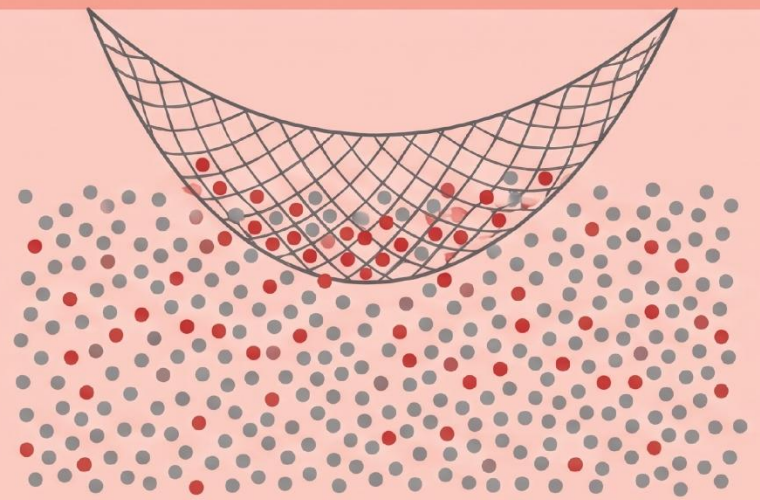
Let's assume that the dispersion is similar to the standard error from DESeq2.





What do we need an adjusted *p*-value?

WITHOUT correction
Raw p-value



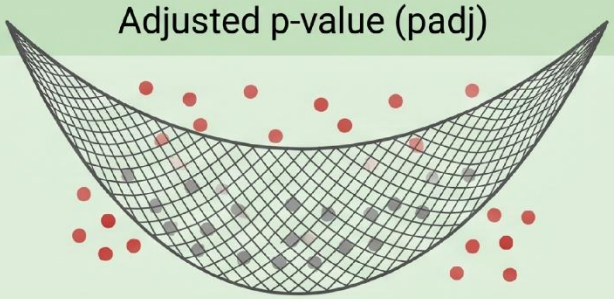
We test 20,000 genes
Using p < 0.05
→ ~1,000 false positives by chance

⚠ Problem: raw p-values include chance hits

VS

Benjamini–Hochberg correction

WITH correction
Adjusted p-value (padj)



Adjusted p-value (padj) = FDR
FDR = False Discovery Rate

padj = 0.03 means **3% false positives** among significant genes

✓ Standard threshold: padj < 0.05
→ At most 5% false positives among significant genes

Simple example

Gene	raw p	padj	significance
CTLA4	0.001	0.008	✓ Significant
CD28	0.03	0.18	✗ Not significant after correction
FOXP3	0.0001	0.001	✓ Significant

CD28 had raw p = 0.03, but after correction it is not significant.
Use padj in RNA-seq — not raw p alone.

DESeq2 pipeline uses Benjamini-Hochberg Correction.

What is a Benjamini-Hochberg (BH) Correction?

Sort all p-values from smallest to largest.

Rank	*p*-value	N / rank	*p*adj
1	0.001	x (5/1)	0.005
2	0.008	x (5/2)	0.020
3	0.039	x (5/3)	0.065
4	0.041	x (5/4)	0.051
5	0.220	x (5/5)	0.220

Let's practice a little. Copy the code into RStudio

```r
# Create the dataframe
df <- data.frame(
  gene = c("TP53", "BRCA1", "MYC", "EGFR", "PTEN"),

  control_1 = c(10, 50, 30, 80, 25),
  control_2 = c(12, 48, 32, 75, 27),
  control_3 = c(9, 52, 28, 83, 24),

  treated_1 = c(20, 55, 90, 70, 30),
  treated_2 = c(22, 53, 85, 72, 29),
  treated_3 = c(18, 57, 95, 68, 31)
)

# Correct the rownames
rownames(df) <- df$gene
df$gene <- NULL

# Mean expression for each condition
mean_control <- rowMeans(df[, c("control_1", "control_2", "control_3")])
mean_treated <- rowMeans(df[, c("treated_1", "treated_2", "treated_3")])

# Fold change and log2 fold change
fold_change <- mean_treated / mean_control
logFC <- log2(fold_change)

df$logFC <- logFC

# Illustrative p-values (replace with actual Wald-test p-values)
df$pvalue <- c(0.001, 0.20, 0.0001, 0.03, 0.50)

# Adjust p-values using Benjamini-Hochberg
df$padj <- p.adjust(df$pvalue, method = "BH")

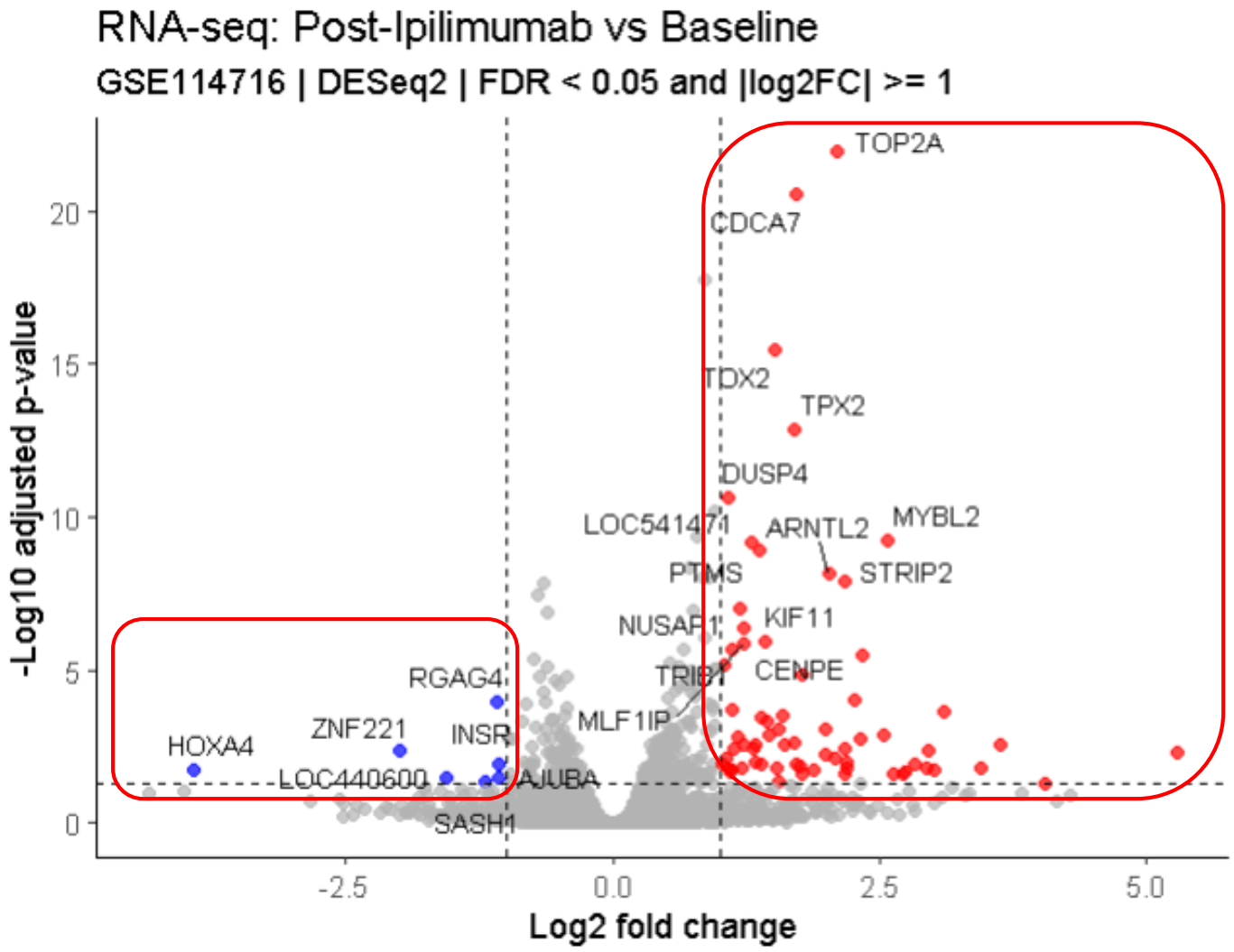
# View results
View(df)
```



Taking a Deeper Look at Gene Functional Analysis

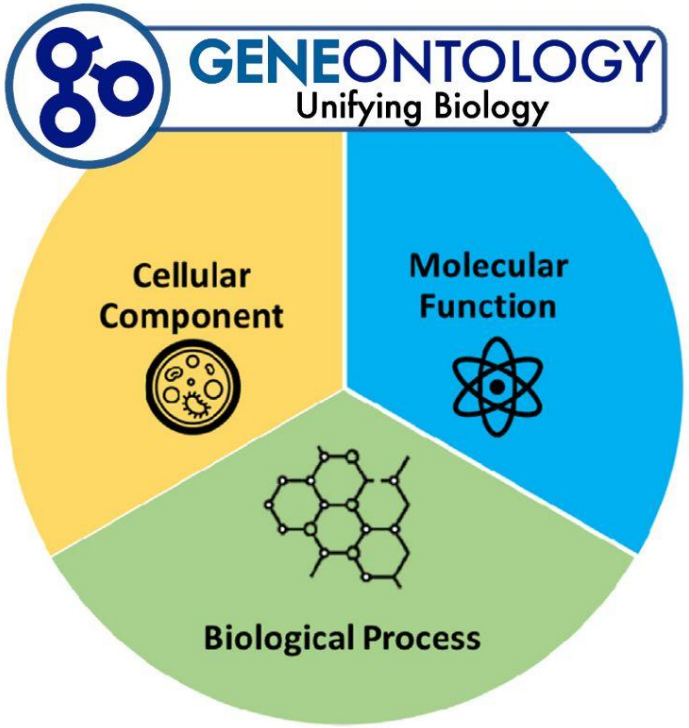
Theory behind Functional Enrichment

Selection of Differentially Expressed Genes



Differentially Expressed Genes (DEGs)

“Using these DEGs, we will ask — do these genes share a common biological function, and is that function being activated or repressed after treatment?”

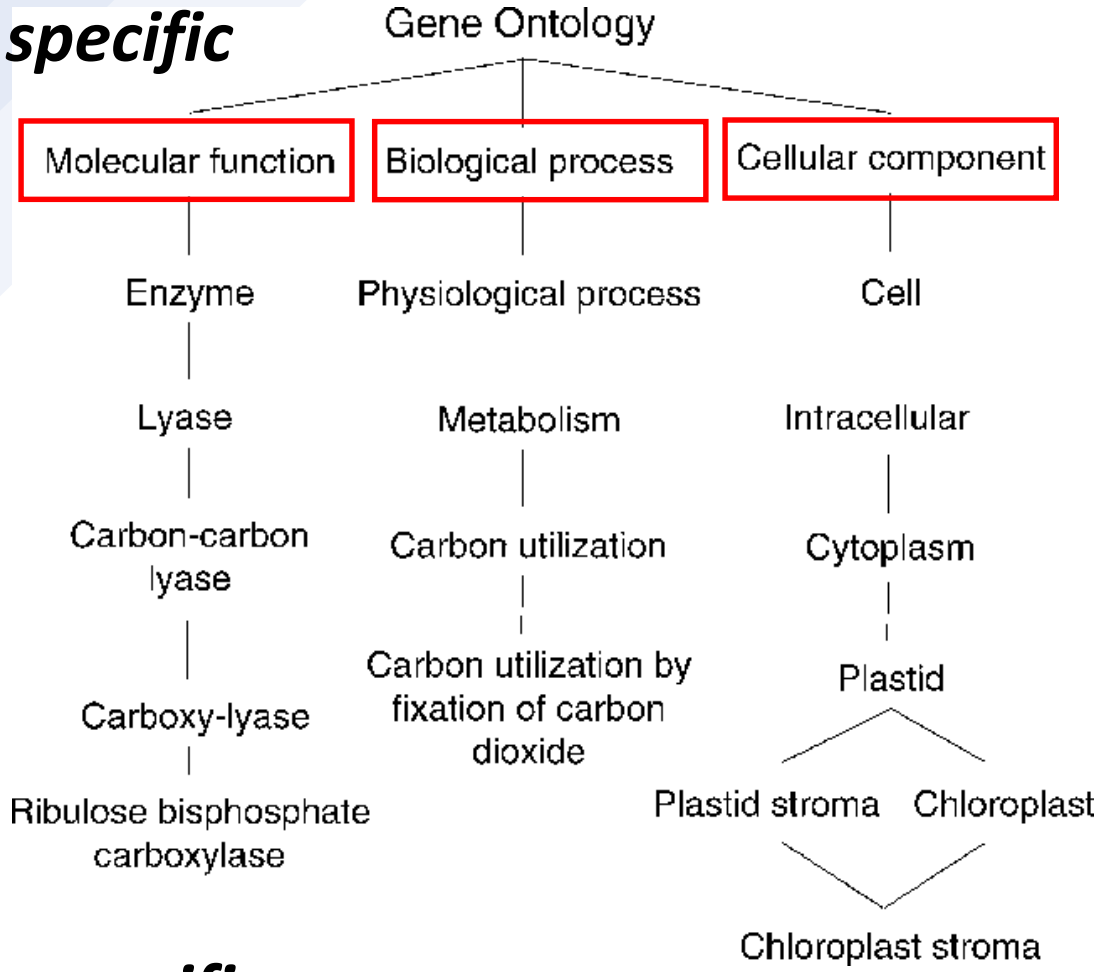


https://doi.org/10.1016/B978-0-323-89775-4.00015-8

Overrepresentation analysis

Hierarchical-based

Less specific



More specific

Gene Ontology: A classification system that groups genes based on their biological functions, cellular location, or molecular role.

- *Objective*: Identify which biological processes, cellular components, or molecular functions are associated with a set of genes.
- *Input*: A list of previously selected genes (e.g., differentially expressed genes).
- *Output*: Enriched GO terms (functions or processes), i.e., more represented in this list of genes than would be expected by chance.

List of all genes from differential gene expression analysis

	Gene symbol	p-val	Differentially expressed	FC
1	*HAO1*	0.0001	Yes	-5.6
2	*RDH16*	0.00045	Yes	-7.6
3	*ALDOB*	0.00009	Yes	-6.7
4	*AKR1C4*	0.0051	No	1.99
5	*ABCB11*	0.67	No	2.5
6	*BAAT*	0.094	No	23.5
7	*SLC2A2*	0.00990	Yes	-2
8	*FCER1G*	0.00673	Yes	6.7
		...		
6532	*OSM*	0.00009	Yes	9
6533	*VEGFA*	0.37	No	8.5
6534	*ZAP70*	0.0034	Yes	7.8

Threshold for DEG* selection:

p-val < 0.05

|FC| > 2

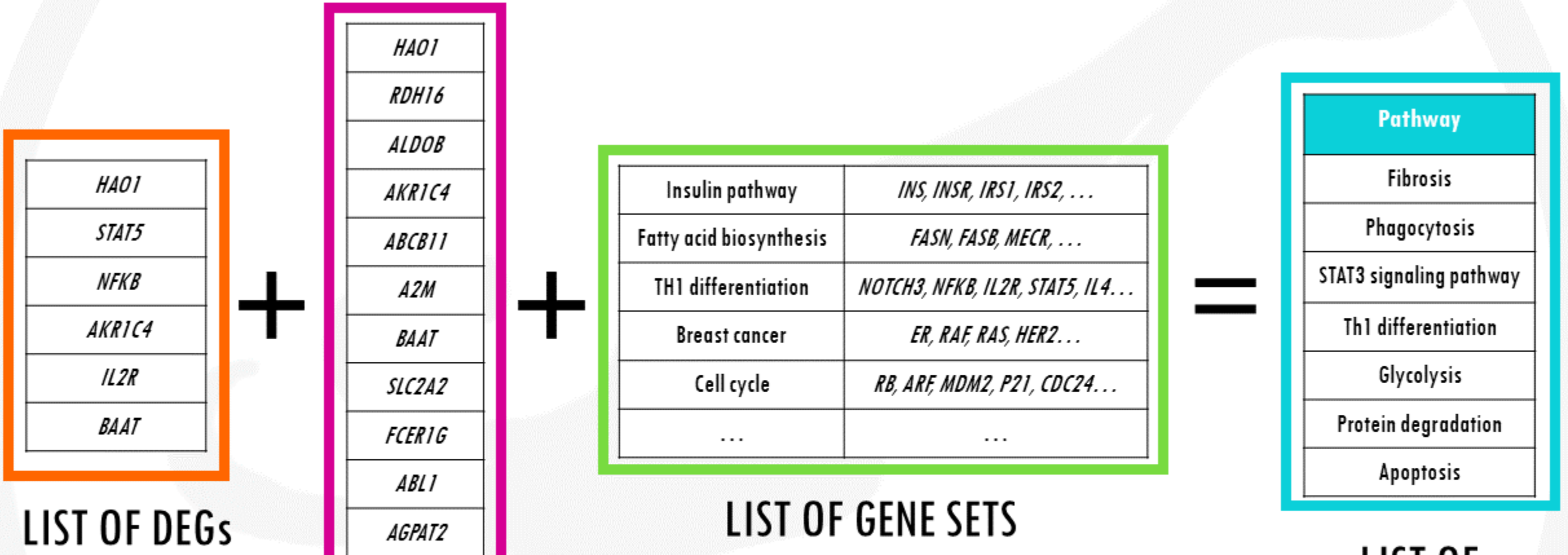
List of DEGs

	Gene symbol
1	*HAO1*
2	*RDH16*
3	*ALDOB*
5	*ABCB11*
6	*BAAT*
7	*SLC2A2*
8	*FCER1G*
	...
800	*OSM*
801	*VEGFA*
802	*ZAP70*

Pathway enrichment analysis

*DEG = Differentially Expressed Gene

PATHWAY ENRICHMENT ANALYSIS



In short… overrepresentation analysis..

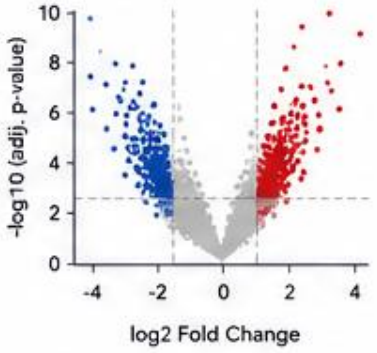
FUNCTIONAL ENRICHMENT (GENE ONTOLOGY)

Works with a list of significant genes

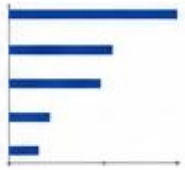
WHAT IS IT?

Tests whether predefined GO terms (BP, MF, CC) are over-represented in a list of differentially expressed genes.

HOW IT WORKS







1 Differential expression analysis

Identify significant genes (e.g., adj. p-value < 0.05, $|\log_2 FC| > 1$)

↓

2 Create gene list

Up-regulated and/or down-regulated genes

↓

3 GO enrichment analysis

Test which GO terms are over-represented in the list (e.g., hypergeometric test)

↓

4 Enriched GO terms

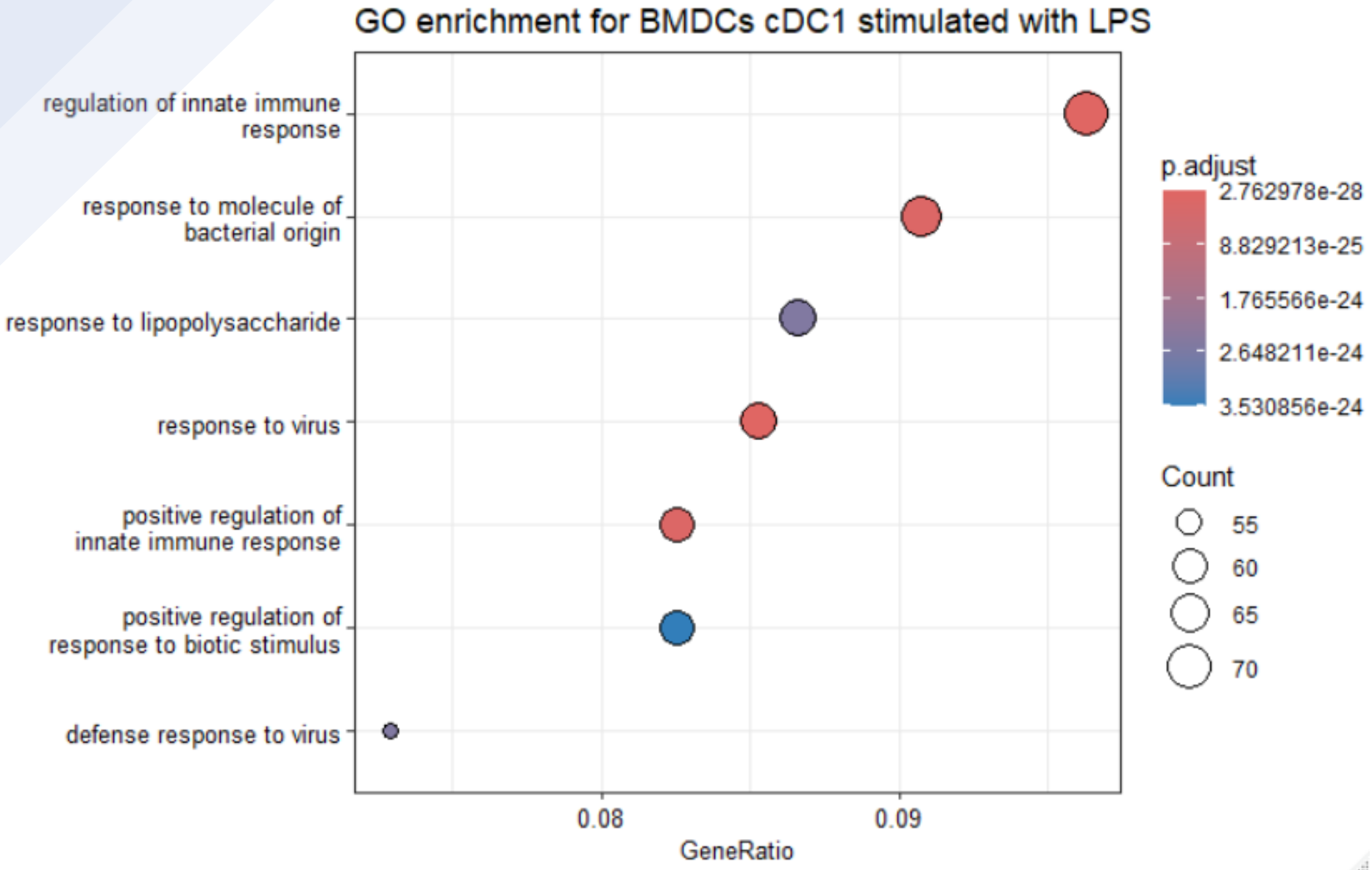
Results reported as GO terms with p-values and multiple testing correction (FDR)

EXAMPLE RESULT

GO Biological Process	Adjusted p-value (FDR)
immune response	1.2×10^{-6}
inflammatory response	3.4×10^{-5}
cell chemotaxis	7.8×10^{-4}
cytokine production	1.6×10^{-3}
leukocyte activation	2.2×10^{-3}
…	

KEY POINTS

- Uses a thresholded gene list
- Sensitive to cut-off choice
- Focuses on "over-representation"
- Good for summarizing large lists of significant genes


GO enrichment for BMDCs cDC1 stimulated with LPS
regulation of innate immune response
response to molecule of bacterial origin
response to lipopolysaccharide
response to virus
positive regulation of innate immune response
positive regulation of response to biotic stimulus
defense response to virus
0.08
0.09
GeneRatio
p.adjust
2.762978e-28
8.829213e-25
1.765566e-24
2.648211e-24
3.530856e-24
Count
55
60
65
70

The problem...

DEGs (p-val < 0.05, |FC| > 2)

	Gene symbol
1	*HAO1*
2	*RDH16*
3	*ALDOB*
4	*AKR1C4*
5	*ABCB11*
6	*BAAT*
7	*SLC2A2*
8	*FCER1G*
	...
6532	*OSM*
6533	*VEGFA*
6534	*ZAP70*

PEA

Pathway	P-val
Glucolysis	0.003

Significant!

DEGs (p-val < 0.01, |FC| > 2)

	Gene symbol
1	*HAO1*
2	*RDH16*
3	*ABCB11*
4	*BAAT*
5	*SLC2A2*
6	*FCER1G*
	...
3001	*VEGFA*
3002	*ZAP70*

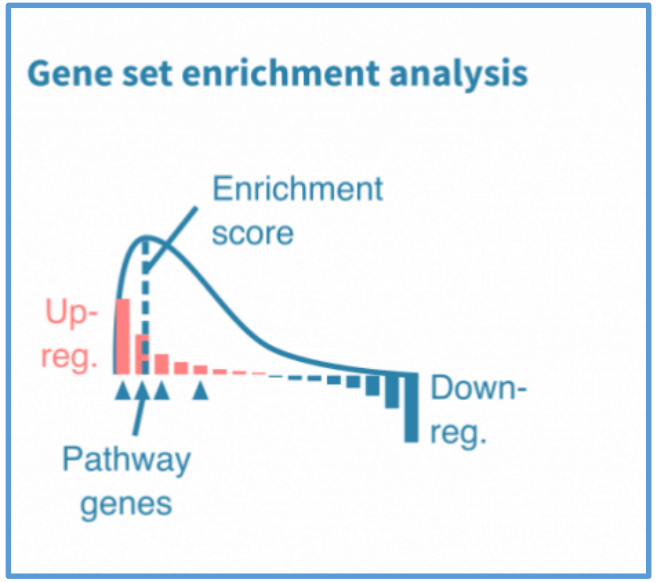
PEA

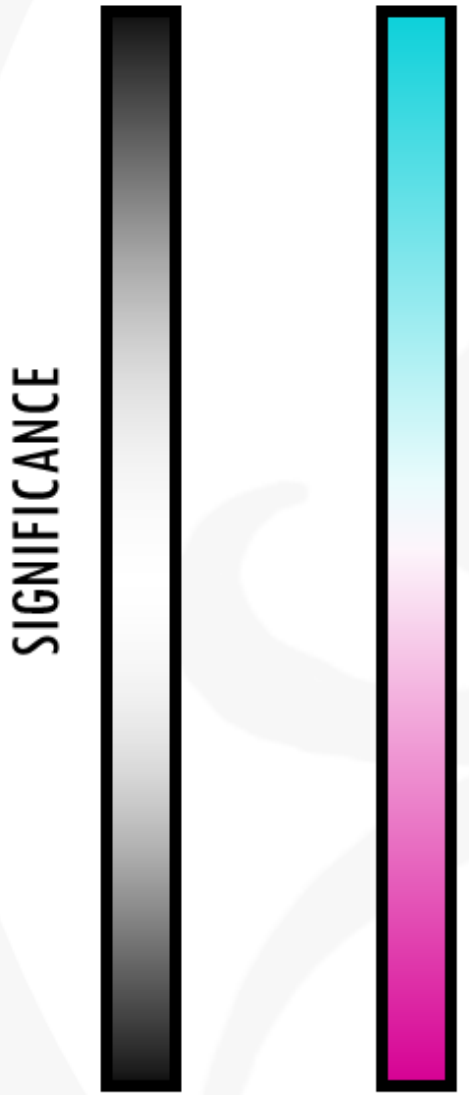
Pathway	P-val
Glucolysis	0.061

Not significant!

RESULTS FOR PATHWAY ENRICHMENT ANALYSIS MAY CHANGE A LOT DEPENDING ON YOUR THRESHOLDS!

What happens to all the other genes that didn’t pass the threshold — even the ones that almost made it? GSEA solves this by using the entire ranked list, so no gene is left behind.





Ranked list

	Gene symbol	Ranking
1	*JAK9*	12.9
2	*RDH16*	11.4
3	*ALDOB*	11.3
5	*AKR1C4*	10.7
6	*ABCB11*	8.6
7	*BAAT*	3.6
8	*SLC2A2*	0.0099
9	*FCER1G*	0.00673
	...	
6532	*MyD88*	-9
6533	*TRIFF*	-9.37
6534	*IRAK1*	-15.6



SIGNIFICANT + UPREGULATED

NON-SIGNIFICANT

SIGNIFICANT + DOWNREGULATED

List of all genes from differential gene expression analysis

	Gene symbol	p-val	Differentially expressed	FC
1	*HAO1*	0.0001	Yes	-5.6
2	*RDH16*	0.00045	Yes	-7.6
3	*ALDOB*	0.00009	Yes	-6.7
4	*AKR1C4*	0.0051	Yes	1.99
5	*ABCB11*	0.67	No	2.5
6	*BAAT*	0.094	No	23.5
7	*SLC2A2*	0.00990	Yes	-2
8	*FCER1G*	0.00673		
		...		
6532	*OSM*	0.00009	Yes	9
6533	*VEGFA*	0.37	No	8.5
6534	*ZAP70*	0.0034	Yes	7.8

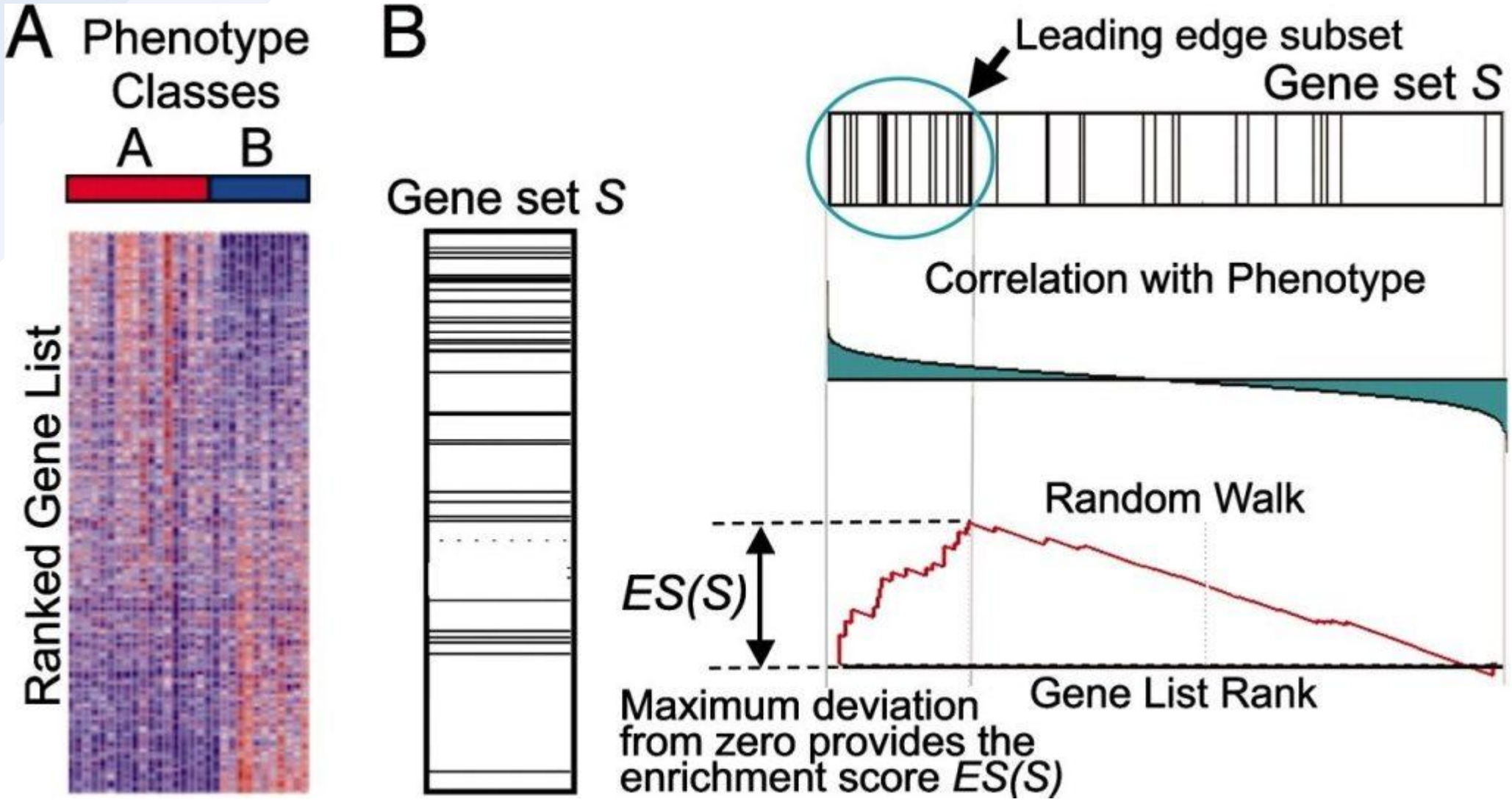
$$RANKING = SIGN(FC) * -\log_{10}(p\text{-value})$$

Ranked list

	Gene symbol	Ranking
1	*HAO1*	-4
2	*RDH16*	-3.34679
3	*ALDOB*	-4.04576
5	*AKR1C4*	2.29243
6	*ABCB11*	0.173925
7	*BAAT*	1.026872
8	*SLC2A2*	-2.00436
9	*FCER1G*	2.171985
6532	*OSM*	4.045757
6533	*VEGFA*	0.431798
6534	*ZAP70*	2.468521

GSEA

In short… Gene Set Enrichment Analysis..



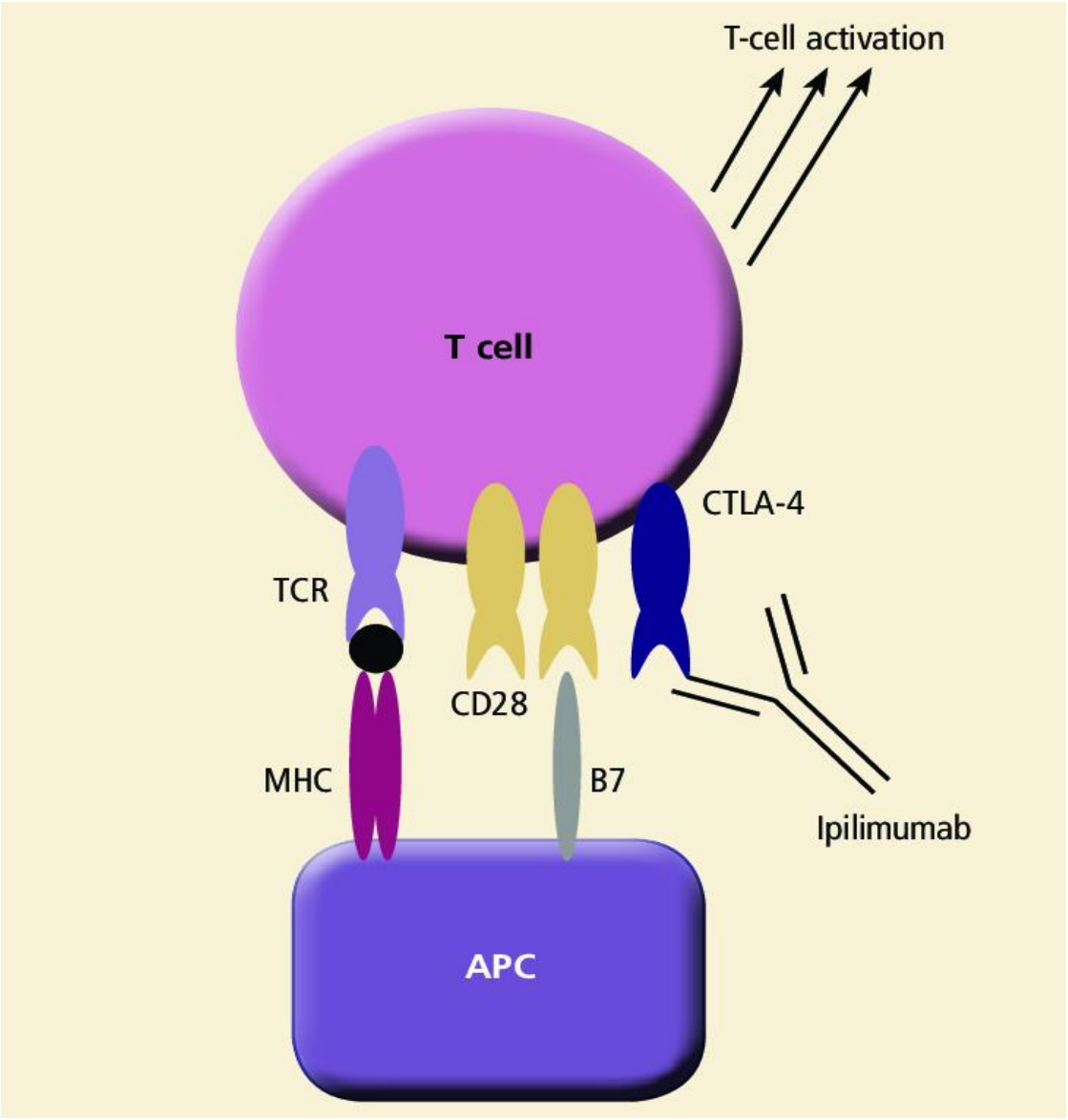


Let's work…

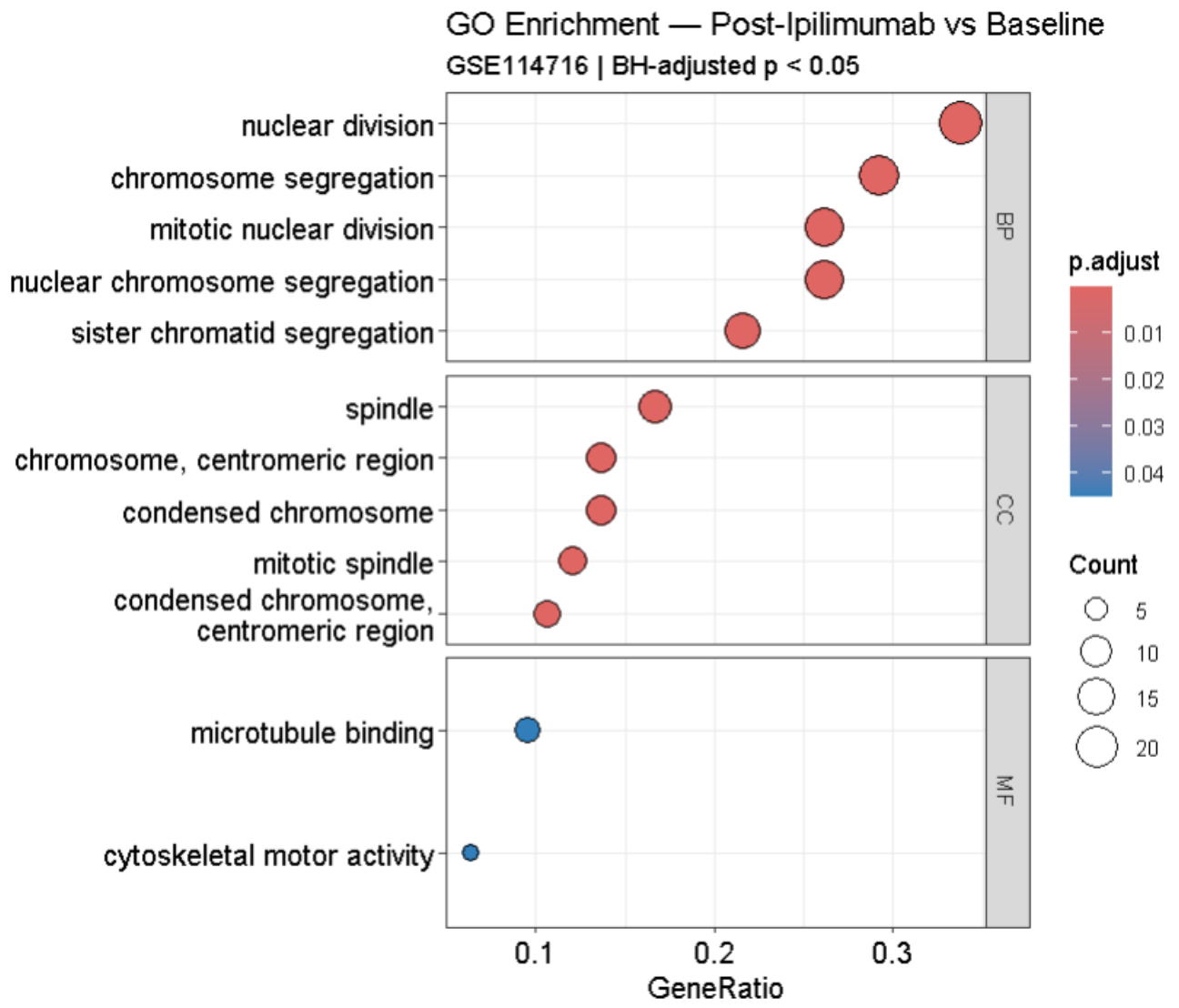
Script Ipilimumab.. Part 2

CTLA4: Cancer Immunity



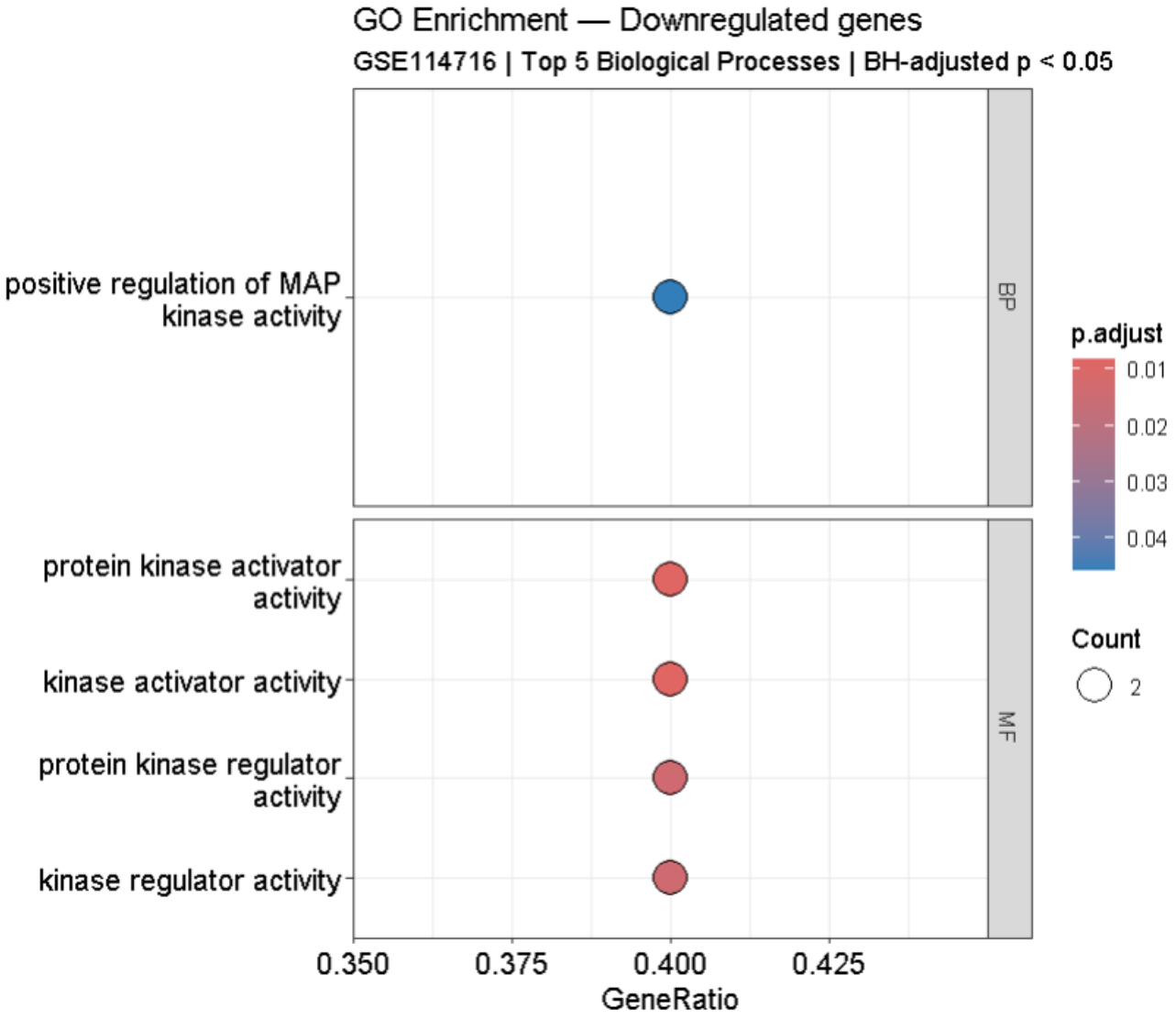


Gene Ontology – upregulated genes



Proliferation / Cellular Division

Gene Ontology – downregulated genes



MAPK activity / Cellular activation

may reflect a resolution of **excessive early signalling**

BUT need to be interpreted with caution – **FEW GENES**

https://www.gsea-msigdb.org/

GSEA
Gene Set Enrichment Analysis

GSEA Home | Downloads | Molecular Signatures Database | Documentation | Contact | Team

UC San Diego

BROAD INSTITUTE

Overview

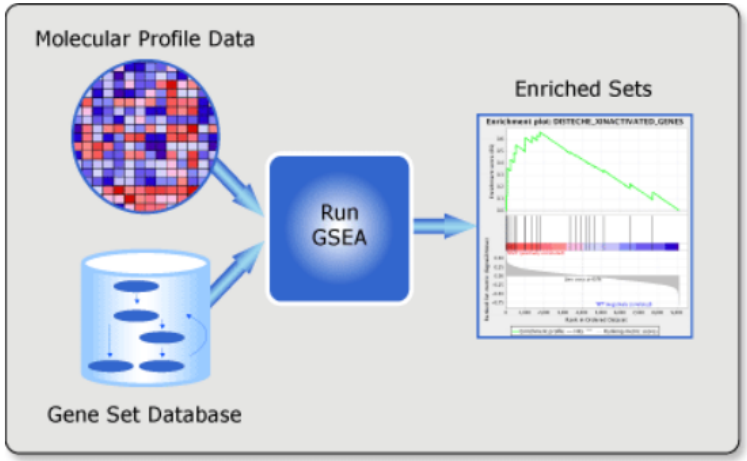
Gene Set Enrichment Analysis (GSEA) is a computational method that determines whether an a priori defined set of genes shows statistically significant, concordant differences between two biological states (e.g. phenotypes).

- **Download** the GSEA software and additional resources to analyze, annotate and interpret enrichment results.
- **Explore the Molecular Signatures Database (MSigDB)**, a collection of annotated gene sets for use with GSEA software.
- **View documentation** describing GSEA and MSigDB.
- View guidelines for **using RNA-seq datasets with GSEA**.

What's New

15-Oct-2026: MSigDB 2026.2 introduces Mouse ontology subcollection M5:MPO derived from the Mammalian Phenotype Ontology database, including new gene sets from general phenotype terms and revised MPT sets related to cancer specific phenotype terms. This release also provides collection updates to KEGG Medicus, GO, Reactome, and WikiPathways. Gene data has been updated to Ensembl 116. See the Human and Mouse release notes for details.

30-Jan-2026: MSigDB 2026.1 introduces the Human C9 collection of computational perturbation signature gene sets. The initial cohort of sets in this collection define the transcriptional signatures of various oncogene dependences. This release also provides collection updates for GO, Reactome,



License Terms

GSEA and MSigDB are available for use under these license terms.

Please register to download the GSEA software, access our web tools, and view the MSigDB gene sets. After registering, you can log in at any time using your email address. Registration is free. Its only purpose is to help us track usage for reports to our funding agencies.

Citing GSEA

To cite your use of the GSEA software, a joint project of UC San Diego and Broad Institute, please reference Subramanian, Tamayo, et al. (2005, PNAS) and Mootha, Lindgren, et al. (2003, Nature Genetics).

Gene Set Enrichment Analysis

GSEA 4.3.2 (Gene set enrichment analysis)

File Downloads Help

Steps in GSEA analysis

- Load data
- Run GSEA
- Leading edge analysis
- Enrichment Map Visualization

Tools

- Run GSEAPreranked
- Collapse Dataset
- Chip2Chip mapping

- Analysis history

GSEA reports

Processes: click 'status' field for results

Name	Status

Show results folder

Home

Steps in GSEA

1. What you need for GSEA

- Expression data set
- Phenotype annotation
- Gene sets – use MSigDB or your own gene sets

2. Run GSEA

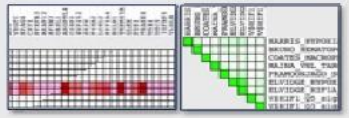
- Start with default parameters
- If you want to collapse probes to genes, specify chip platform

3. View results



4. Leading edge analysis

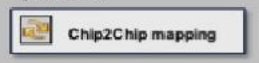
- Leading edge finds genes driving enrichment results



Gene Set Tools

Chip2Chip mapping

- Convert gene sets between platforms



Explore MSigDB gene sets

- See the online tools and data at www.msigdb.org
- Search the database of thousands of gene sets
- Browse the gene sets by name
- Find overlapping gene sets
- Export gene sets

Getting Help

GSEA web site:
www.gsea-msigdb.org

Contact the GSEA team:
gsea-msigdb.org/gsea/contact.jsp

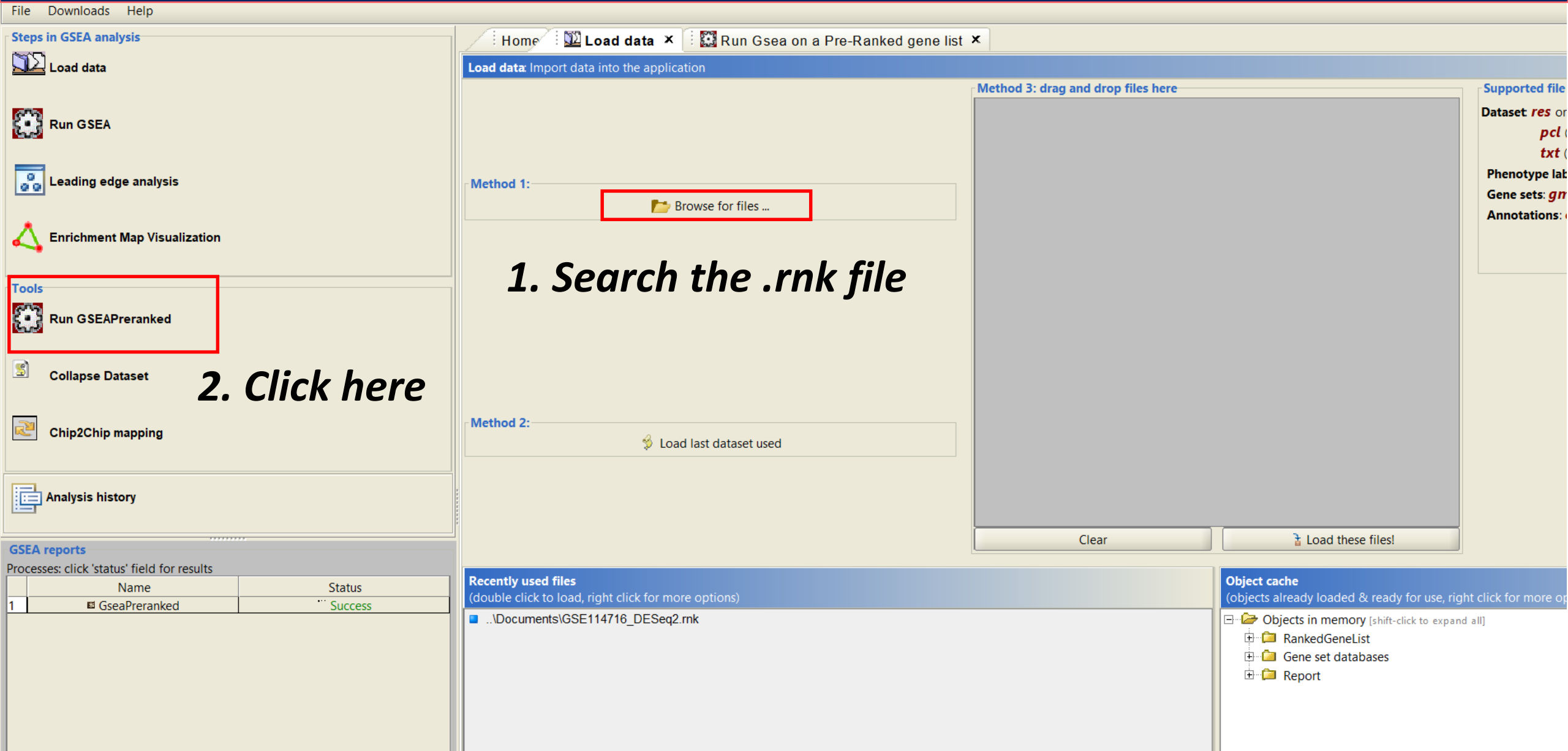
BROAD INSTITUTE

UC San Diego

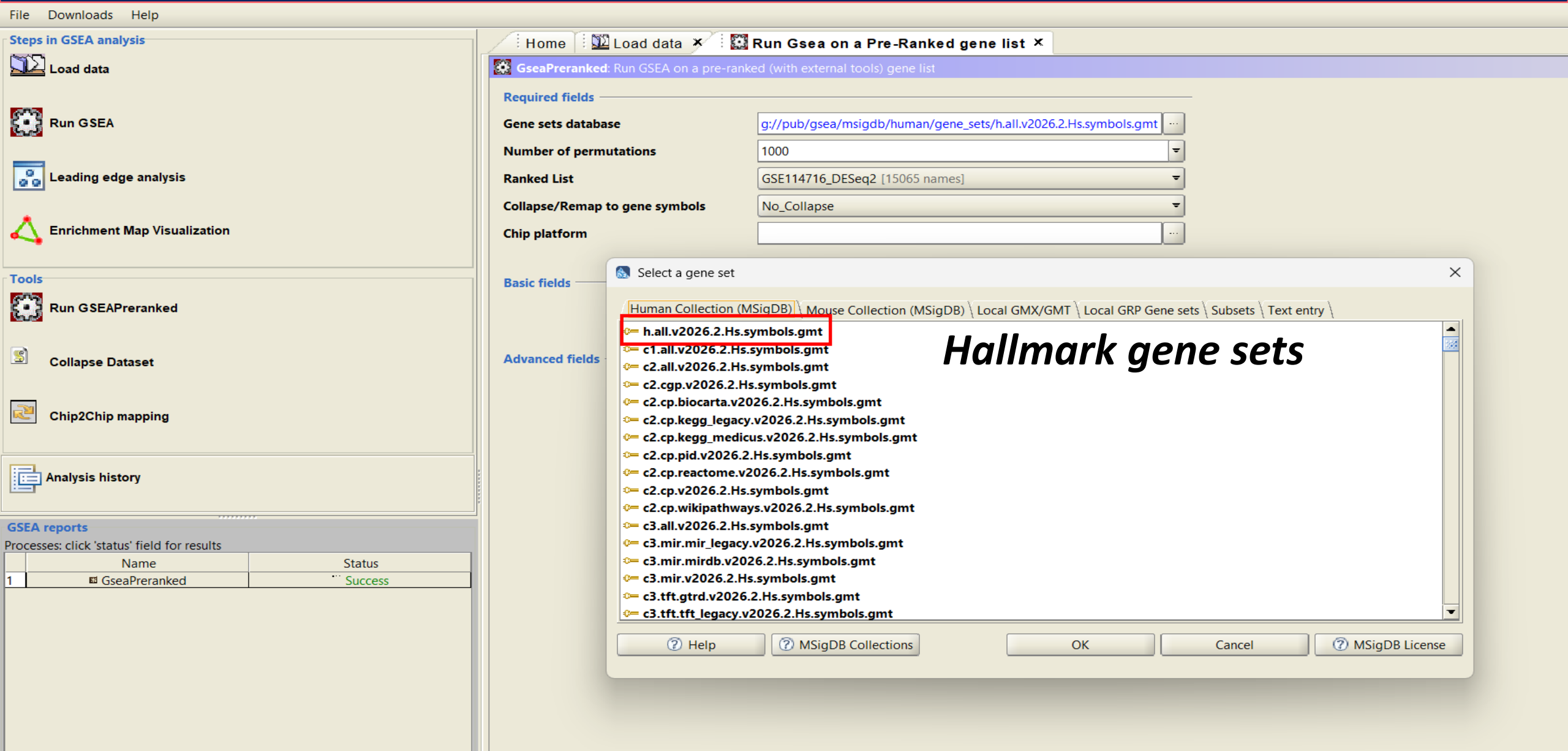
14:11:59

31M of 248M

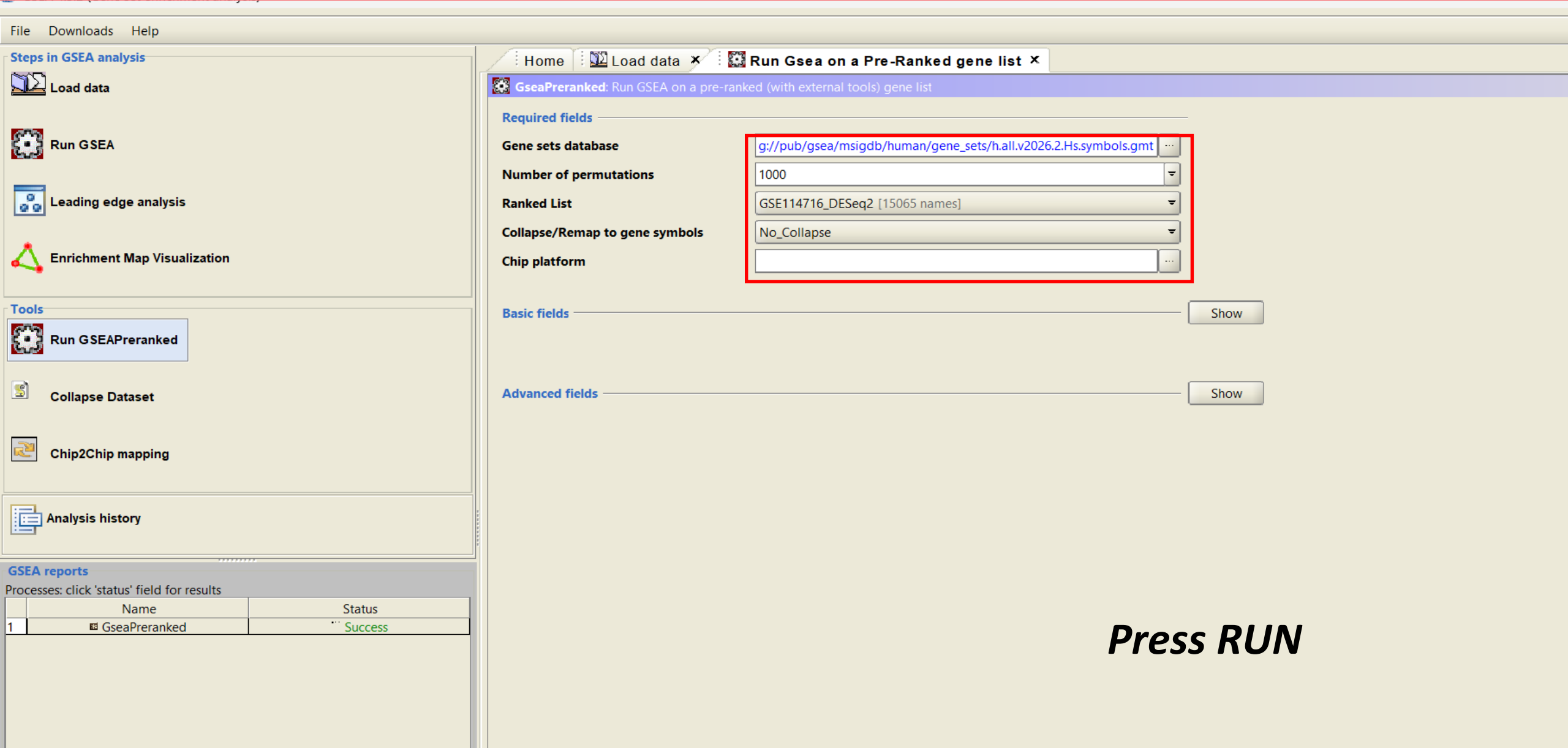
Gene Set Enrichment Analysis



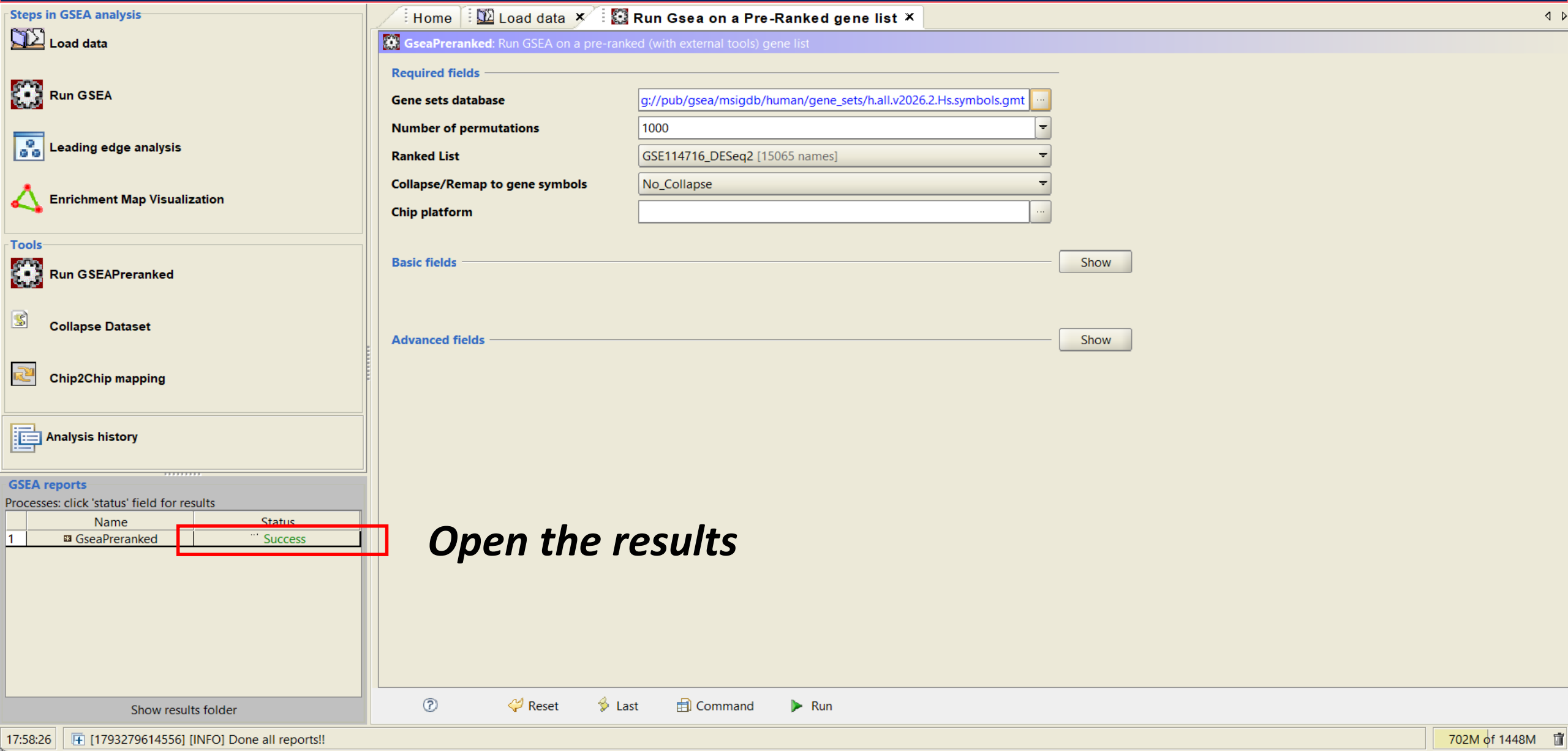
Gene Set Enrichment Analysis



Gene Set Enrichment Analysis



Gene Set Enrichment Analysis



Open the results

GSEA Report for Dataset GSE114716_DESeq2

Enrichment in phenotype: na

- 38 / 50 gene sets are upregulated in phenotype **na_pos**
- 27 gene sets are significant at FDR < 25%
- 10 gene sets are significantly enriched at nominal pvalue < 1%
- 16 gene sets are significantly enriched at nominal pvalue < 5%
- Snapshot of enrichment results
- Detailed enrichment results in html format
- Detailed enrichment results in TSV format (tab delimited text)
- Guide to interpret results

Upregulated gene sets

Enrichment in phenotype: na

- 12 / 50 gene sets are upregulated in phenotype **na_neg**
- 2 gene sets are significantly enriched at FDR < 25%
- 0 gene sets are significantly enriched at nominal pvalue < 1%
- 2 gene sets are significantly enriched at nominal pvalue < 5%
- Snapshot of enrichment results
- Detailed enrichment results in html format
- Detailed enrichment results in TSV format (tab delimited text)
- Guide to interpret results

Downregulated gene sets

Dataset details

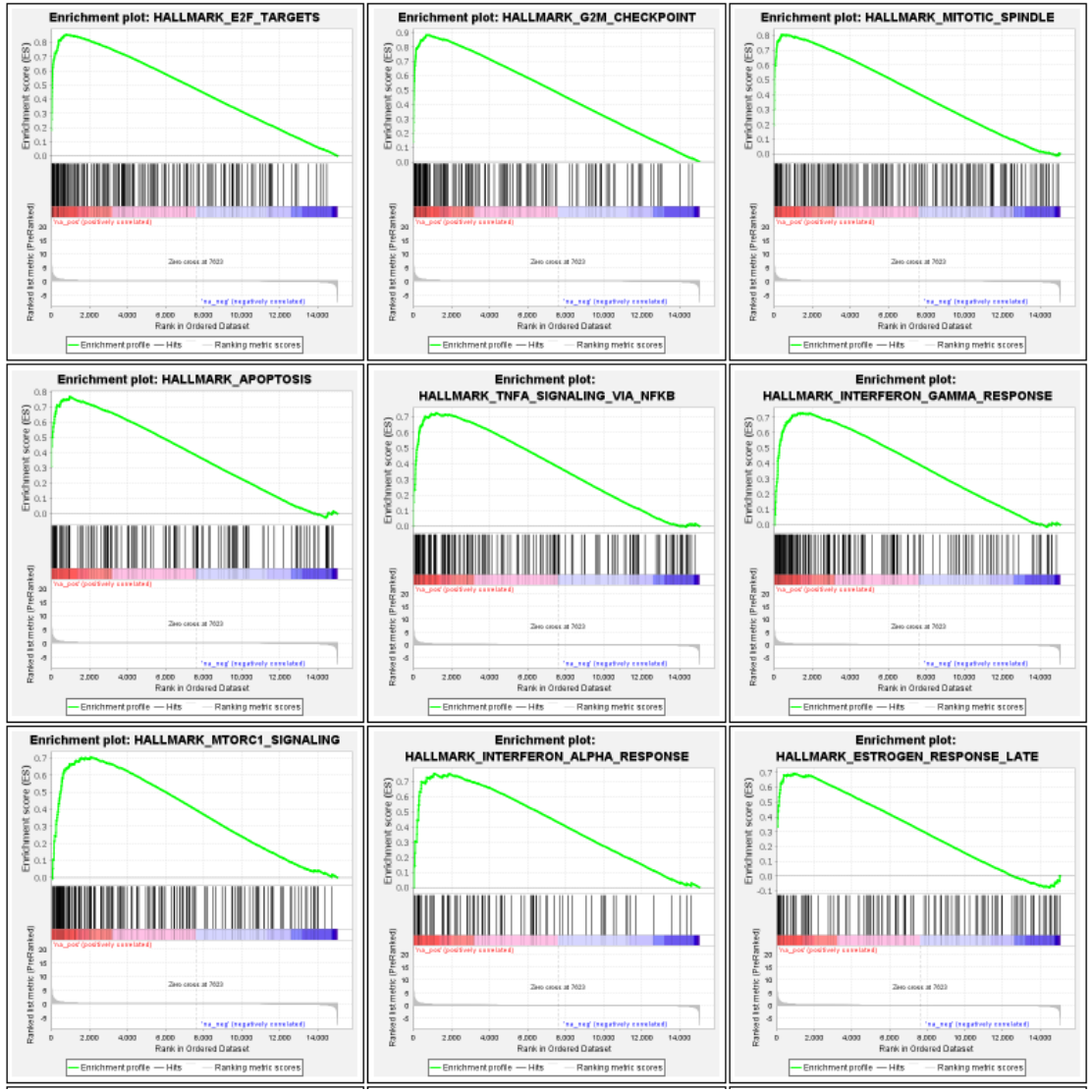
- The dataset has 15065 features (genes)
- No probe set => gene symbol collapsing was requested, so all 15065 features were used

Gene set details

- Gene set size filters (min=15, max=500) resulted in filtering out 0 / 50 gene sets
- The remaining 50 gene sets were used in the analysis

Upregulated Gene Sets

Table: Snapshot of enrichment results



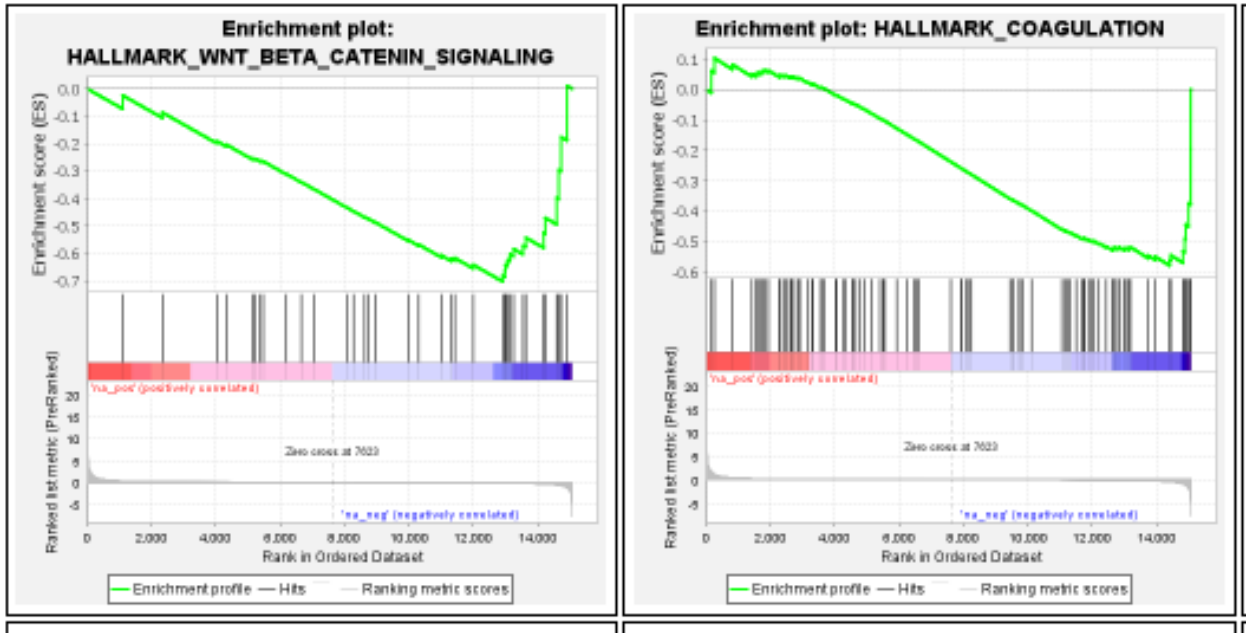
After Ipilimumab

Proliferation / Cellular Division

Inflammatory pathways

Downregulated Gene Sets

After Ipilimumab



May reflect less stemness in these cells / High activation